BIO 102

VIRUS

- **CHARACTERISTICS OF VIRUS**
- **METHODS OF IDENTIFICATION OF VIRUS**
- **CLASSIFICATION OF VIRUS**
- **DISEASES TRANSMITTED BY VIRUS**
- **ROUTES OF VIRAL DISEASE TRANSMISSION**

A virus is a very small infectious agent that can only reproduce inside the living cells of a host organism. Virus lacks the structures needed for metabolism, such as the ability to produce energy or carry out independent biochemical reactions, and it cannot reproduce on its own. A virus is made up of genetic material, either DNA or RNA, enclosed in a protein coat called a capsid, and in some cases, an additional outer lipid envelope.

1. CHARACTERISTICS OF VIRUSES

i. **Virus are acellular in nature**: Viruses are considered acellular because they do not have a cellular structure. This means they lack the typical components of a cell, such as organelles, ribosomes, a plasma membrane etc, rather virus exists as a virion when it's outside of a host cell. The virion is the infectious form of the virus enclosed within a protective protein coat called capsid.

ii. **Contain genetic materials**: Each virus contains either DNA or RNA as its genetic material, but never both. The nucleic acid of a virus, which may be either DNA or RNA, can exist as single-stranded or double-stranded molecules. Examples include single-stranded DNA (ssDNA), double-stranded DNA (dsDNA), single-stranded RNA (ssRNA), and double-stranded RNA (dsRNA). The type and structure of the viral nucleic acid greatly influence the virus's replication strategy, genetic stability, and interaction with host cells.

iii. **Dependence on Host Cells**: Viruses are obligate intracellular parasites that rely entirely on a host cell to reproduce and perform metabolic functions. Therefore they cannot reproduce or carry out metabolic activities without infecting a host cell. Viruses are extremely small, usually ranging from 20 to 300 nanometers, and are much smaller than most pathogens.

iv. **Protective Protein Coat**: The viral genetic material is surrounded by a protective protein coat called the **capsid**. The capsid is the outer protein shell of a virus. It surrounds and protects the viral genetic material (DNA or RNA) from damage.

v. **Presence of a lipid Envelope**: Some virus possess a lipid envelope derived from the host cell membrane that surrounds the capsid. A viral envelope is a lipid membrane that surrounds the capsid of certain viruses. It is typically derived from the host cell's membrane during viral replication.

vi. **Host Specificity**: Viruses exhibit specificity for their hosts, usually infecting only particular species or cell types based on receptor compatibility. Viruses can only infect certain hosts or specific types of cells because they need to bind to particular receptors on the surface of those cells. This virus has proteins that fit only certain receptors on host cells. Examples: HIV (Human Immunodeficiency Virus): Infects mostly human immune cells that have the CD4 receptor and a co-receptor (CCR5 or CXCR4).

Influenza Virus infects respiratory tract cells because it binds to sialic acid receptors on those cells.
Rabies Virus: Infects nerve cells because it binds to receptors found on neuronal cells.

vii. **Lack of Metabolic Activity**: Viruses do not perform independent metabolism; they neither respire nor grow on their own.

viii. **High Mutation Rate:** Viruses replicate very quickly, often producing millions of copies in a short time, especially among RNA viruses, allowing viruses to evolve quickly and adapt to changing environments. The more the replication, the higher the chances of mutation

2. METHODS OF IDENTIFICATION OF VIRUSES

Identifying viruses is a crucial step in diagnosing viral infections, conducting epidemiological surveillance, and developing targeted treatment strategies. Viruses can be identified by detecting the whole virus, its components, or the host's immune response. The main methods are:

1. **Virus Isolation and Culture**: This involves growing viruses in cell cultures, eggs, or animals and observing cytopathic effects (Cell injury or death). It's the gold standard but time-consuming (e.g., Herpesvirus, Influenza).
2. **Electron Microscopy (EM):** Electron microscopy is the visualization of virus particles to determine size and shape, useful for rapid identification of large or unknown viruses (e.g., Poxvirus, Adenovirus).
3. **Detection of Viral Antigens**: This method involve identifying viral proteins in samples using assays like immunofluorescence and ELISA for early diagnosis (e.g., Influenza, Respiratory Syncytial Virus-RSV).
4. **Detection of Viral Nucleic Acids**: Detection of Viral Nucleic Acids using molecular techniques such as PCR and next-generation sequencing to detect viral DNA or RNA with high sensitivity (e.g., HIV, SARS-CoV-2).
5. **Serological Methods**: Serological Methods detects antibodies produced by the host in response to infection to diagnose current or past infections (Commonly seen in Measles, Rubella).
6. **Rapid Diagnostic Tests (RDTs):** These are point-of-care tools that provide quick results, typically within minutes. These immunochromatographic assays detect either viral antigens or antibodies and are highly useful in field conditions and emergency settings. Commonly used for HIV, hepatitis, COVID-19, malaria etc diagnosis.

3. CLASSIFICATION OF VIRUSES

Viruses are classified based on several criteria these includes:

1. CLASSIFICATION OF VIRUSES BASED ON THE TYPE OF GENETIC MATERIAL AND METHOD OF REPLICATION

This classification is called the Baltimore classification system. It categorizes viruses according to the type of their genetic material and their method of replication. This system, proposed by David Baltimore, divides viruses into seven groups based on whether their genome consists of DNA or RNA, whether it is single-stranded or double-stranded, and how the virus synthesizes messenger RNA (mRNA) during infection. They are classified as:

Group I: Double-Stranded DNA Viruses (dsDNA)
These viruses have a double-stranded DNA genome. They typically replicate inside the host cell nucleus using the host's DNA-dependent RNA polymerase to produce viral mRNA. Examples include adenoviruses and herpesviruses.

Group II: Single-Stranded DNA Viruses (ssDNA)
Viruses in this group carry single-stranded DNA genomes. Upon entering the host cell, their ssDNA is converted into double-stranded DNA before transcription into mRNA occurs. Parvoviruses are typical members of this group.

Group III: Double-Stranded RNA Viruses (dsRNA)
These viruses contain double-stranded RNA genomes. They carry their own RNA-dependent RNA polymerase to transcribe the negative strand of their RNA into positive-sense mRNA. Eg. Reoviruses.

Group IV: Single-Stranded Positive-Sense RNA Viruses (+ssRNA)
These viruses have single-stranded RNA genomes that act directly as mRNA. This means their RNA can be immediately translated by the host's ribosomes to produce viral proteins. Examples include poliovirus and hepatitis C virus.

Group V: Single-Stranded Negative-Sense RNA Viruses (-ssRNA)
Viruses in this group contain single-stranded RNA genomes complementary to mRNA. They must carry an RNA-dependent RNA polymerase enzyme within their virions to transcribe their genome into positive-sense mRNA after infection. Influenza virus and rabies virus are notable members.

Group VI: Single-Stranded RNA Viruses with Reverse Transcriptase (ssRNA-RT)
These are retroviruses that contain single-stranded RNA genomes but replicate through a DNA intermediate. They use reverse transcriptase to convert their RNA into DNA, which integrates into the host genome. Eg. HIV.

Group VII: Double-Stranded DNA Viruses with Reverse Transcriptase (dsDNA-RT)
Viruses in this group have double-stranded DNA genomes but replicate via an RNA intermediate. They use reverse transcription during their replication cycle. Hepatitis B virus belongs to this category.

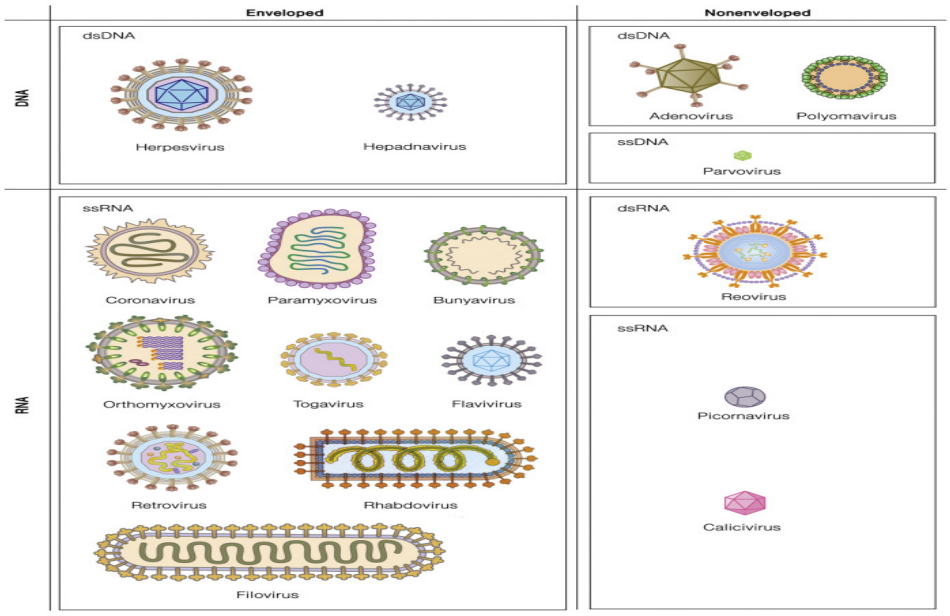


Fig. 1. Classification of Virus

2. CLASSIFICATION BASED ON PRESENCE OR ABSENCE OF A LIPID ENVELOPE

This is a fatty outer layer that surrounds some viruses. It is derived from host cell membrane. This characteristic affects how viruses are transmitted, how they enter host cells, and how they are protected in the external environment.

i. **Enveloped viruses**: These are viruses surrounded by a lipid membrane derived from host cells (e.g., Influenza virus, Herpes simplex virus (causes cold sores, genital herpes).

ii.**Non-enveloped (naked) viruses**: These are the viruses that lack the lipid envelope. Non-enveloped viruses are often transmitted via the fecal-oral route or through contact with contaminated surfaces. (e.g., Adenovirus, Poliovirus-causative agent of polio myelitis)

3. VIRUS CLASSIFICATION BASED ON HOST SPECIFICITY

Viruses are incredibly diverse in terms of their hosts, and one of the most fundamental ways to classify viruses is based on the type of organisms they infect. Host specificity refers to the specific type of organism or cell that a virus is capable of infecting. This characteristic is crucial for understanding viral transmission, pathogenicity, and the development of treatments and vaccines. Viruses can be broadly classified into three categories based on their host specificity:

i. **Animal Viruses**

Animal viruses are those viruses that infect animals and humans. These viruses can cause a wide range of diseases, from mild illnesses to severe and life-threatening conditions. Animal viruses are highly adapted to their specific host cells, often targeting particular tissues or

organs. For example, the Human Immunodeficiency Virus (HIV) targets immune cells such as CD4 cells, leading to acquired immunodeficiency syndrome (AIDS), while Influenza viruses affect the respiratory tract. These viruses may be transmitted through direct contact, airborne particles, or bodily fluids, depending on the nature of the virus.

ii. **Plant Viruses**

Plant viruses specifically infect plants, causing diseases that can severely affect crop yields and agricultural productivity. These viruses do not infect animals or humans but instead are specialized to infect plant cells. Plant viruses are usually transmitted by insects (such as aphids), through contaminated tools, or by infected seeds. One well-known example of a plant virus is the Tobacco Mosaic Virus (TMV), which infects tobacco plants and other species, causing characteristic mottling and distortion of leaves. Other examples include Potato virus, Tomato yellow leaf curl virus, Cassava mosaic virus etc. Plant viruses often lack specialized receptors for cell entry. They rely on mechanical transmission, such as through wounds in plant tissues or vector organisms. Control measures often involve genetic resistance in plants, crop rotation, and vector control to minimize transmission.

iii. **Bacteriophages**

Bacteriophages are viruses that specifically infect bacteria. These viruses are highly specific to their bacterial hosts and can be used to target particular bacterial strains. Bacteriophages have an important ecological role in controlling bacterial populations, and they are considered natural regulators of bacterial growth. For example, the T4 bacteriophage infects *Escherichia coli*.

Some viral diseases and Their Causative Viruses

1. HIV/AIDS - Human Immunodeficiency Virus (HIV)
2. Influenza (Flu) -Influenza virus
3. Hepatitis B -Hepatitis B virus (HBV)
4. Hepatitis C -Hepatitis C virus (HCV)
5. COVID-19-SARS-CoV-2 (Severe Acute Respiratory Syndrome Coronavirus 2)
6. Measles -Measles virus
7. **Chickenpox (Varicella) -Varicella-zoster virus**
8. Polio (Poliomyelitis**)** - Poliovirus
9. Dengue Fever -Dengue virus
10. Zika Virus Disease -Zika virus
11. Ebola Virus Disease - Ebola virus
12. Rabies -Rabies virus
13. Smallpox -Variola virus
14. Norovirus Infection -Norovirus
15. Human Papillomavirus (HPV) Infection -Human papillomavirus
16. Yellow Fever -Yellow fever virus
17. Herpes Simplex Infection - Herpes simplex virus (HSV)
18. Lassa Fever -Lassa virus
19. Marburg Virus Disease - Marburg virus

PATHWAYS (ROUTES) OF VIRAL DISEASE TRANSMISSION

Viral diseases can spread through various routes depending on the virus and its characteristics.

1. Airborne Transmission

Airborne transmission occurs when viruses spread through respiratory droplets or aerosols released when an infected person coughs, sneezes, talks, or breathes. These particles can be inhaled by others, especially in crowded or poorly ventilated areas. Some viruses can remain suspended in the air for long periods, increasing the risk of infection. Examples of airborne viral diseases include Influenza, COVID-19, Measles, and Chickenpox.

2. Direct Contact Transmission

Direct contact transmission occurs when a person becomes infected through physical contact with an infected individual or exposure to contaminated bodily fluids such as blood, saliva, or mucus. This type of transmission can occur through touching, kissing, sexual contact, or contact with infected wounds and secretions. Examples of viruses spread through direct contact include HIV/AIDS, Herpes Simplex virus (HSV), Ebola Virus Disease, Hepatitis B, Hepatitis C, and Human Papillomavirus Infection (HPV).

3. Fecal-Oral Transmission

Fecal-oral transmission occurs when a person consumes food or water contaminated with the feces of an infected individual. This mode of transmission is common in areas with poor sanitation, inadequate hygiene, and unsafe water supplies. Viruses spread through this route infect the digestive system and can easily spread within communities. Examples include Hepatitis A, Rotavirus Infection, Poliomyelitis, and Norovirus Infection.

4. Vector-Borne Transmission

Vector-borne transmission occurs when viruses are spread through vectors such as mosquitoes, ticks, or other non-arthropod vectors such as animals that carry infectious agents from an infected host to a healthy individual. Transmission usually occurs through bites, during which the virus enters the bloodstream of the new host. This mode of transmission is common in tropical and subtropical regions where vectors thrive. Examples include Dengue Fever, Zika Virus Disease, Yellow Fever, West Nile Fever all these are transmitted by arthropod vectors and Rabies, which is transmitted through bites from infected animals(dogs, bats), lassa fever virus, Hantavirus pulmonary syndrome transmitted through contact with droppings, urine, or saliva of infected rodents and through contact with food or household items contaminated by infected rodents respectively.

7. Sexual Transmission

Sexual transmission occurs when viruses are spread through sexual contact, including vaginal, anal, or oral intercourse. The viruses are commonly transmitted through bodily fluids such as semen, vaginal secretions, blood, or through direct contact with infected skin or mucous membranes. This mode of transmission plays a major role in the spread of several

viral infections. Examples include HIV/AIDS, Human Papillomavirus Infection (HPV), and Herpes Simplex (HSV).

8. Dermatologic Viral Infections

Some viruses primarily affect the skin, leading to rashes, lesions, blisters, and other dermatological conditions. These viral infections may spread through direct contact, respiratory droplets, or contaminated objects. One example is Chickenpox, caused by the Varicella-zoster virus, which produces an itchy skin rash and fluid-filled blisters. Another example is Herpes Simplex (HSV), which is responsible for cold sores and genital herpes. Molluscum Contagiosum is another skin infection that causes small, raised bumps on the skin, especially in children.

MODE OF RESPIRATION IN PLANTS AND ANIMALS

Respiration is a fundamental metabolic process through which living organisms release energy from organic compounds to sustain cellular activities. It involves a series of biochemical reactions that oxidize food substances, particularly carbohydrates, resulting in the production of adenosine triphosphate (ATP), the primary energy carrier in living cells. Respiration occurs in all living organisms, including plants and animals, and is essential for growth, development, movement, reproduction, biosynthesis, active transport, and maintenance of cellular homeostasis.

Although respiration is often associated with breathing, the two processes are distinct. Breathing refers to the physical exchange of gases between an organism and its environment, whereas respiration refers to the intracellular biochemical reactions that release energy from food molecules. In both plants and animals, respiration may occur aerobically in the presence of oxygen or anaerobically when oxygen is absent or insufficient.

Respiration in Plants

Respiration in plants is the process by which organic compounds synthesized during photosynthesis are oxidized to release energy required for physiological and metabolic activities. Unlike photosynthesis, which occurs only in green tissues and in the presence of light, respiration occurs continuously in all living plant cells throughout the day and night.

The primary substrate for plant respiration is glucose, although lipids, proteins, and organic acids may also serve as respiratory substrates under certain physiological conditions. The energy generated through respiration supports cellular processes such as nutrient uptake, biosynthesis, cell division, growth, flowering, fruit development, and seed germination.

Aerobic Respiration in Plants

Aerobic respiration is the complete oxidation of glucose in the presence of oxygen. It is the predominant respiratory pathway in most plant tissues and yields a substantial amount of ATP.

The overall reaction is represented by:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy (ATP)}$

Aerobic respiration occurs through three major stages:

1. Glycolysis

Glycolysis occurs in the cytoplasm and involves the enzymatic breakdown of one molecule of glucose into two molecules of pyruvate. This process produces a small amount of ATP and reduced nicotinamide adenine dinucleotide (NADH).

2. Krebs Cycle (Citric Acid Cycle)

The pyruvate molecules produced during glycolysis are transported into the mitochondria and converted into acetyl coenzyme A (Acetyl-CoA). Acetyl-CoA enters the Krebs cycle, where complete oxidation occurs, releasing carbon dioxide and generating NADH, $FADH_2$, and ATP.

3. Electron Transport Chain and Oxidative Phosphorylation

Electrons carried by NADH and $FADH_2$ are transferred through a series of electron carriers located in the inner mitochondrial membrane. Oxygen serves as the final electron acceptor, resulting in the formation of water. The energy released drives ATP synthesis through oxidative phosphorylation.

Examples of tissues exhibiting high rates of aerobic respiration include germinating seeds, meristematic tissues, developing fruits, and actively growing roots and shoots.

Anaerobic Respiration in Plants

Anaerobic respiration occurs when oxygen availability is limited. Under such conditions, glucose undergoes partial oxidation through fermentation pathways.

In plants and yeast, anaerobic respiration results in the formation of ethanol and carbon dioxide:

$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + \text{Energy}$

This process yields significantly less ATP than aerobic respiration because glucose is not completely oxidized. Anaerobic respiration commonly occurs in flooded soils, waterlogged environments, and certain storage tissues where oxygen diffusion is restricted.

Gaseous Exchange in Plants

Since plants lack specialized respiratory organs, gaseous exchange occurs by simple diffusion through various structures(Stomata, Lenticels, Root hairs and intracellular spaces).

<u>Stomata</u>

Stomata are microscopic pores found primarily on the epidermis of leaves. They regulate the diffusion of oxygen, carbon dioxide, and water vapour between the plant and its environment.

Lenticels

Lenticels are openings present in woody stems and roots that facilitate gaseous exchange between internal tissues and the atmosphere.

Root Hairs and Intercellular Spaces

Roots absorb oxygen from air spaces within the soil through root hairs and interconnected intercellular spaces.

The efficiency of gaseous exchange significantly influences respiratory activity in plants.

Respiration in Animals

Respiration in animals is the process by which organic molecules are oxidized to release energy for physiological functions. Animals require continuous energy supplies to support locomotion, neural activities, thermoregulation, growth, reproduction, and maintenance of internal homeostasis.

Unlike plants, animals possess specialized respiratory systems that facilitate efficient oxygen uptake and carbon dioxide elimination.

Aerobic Respiration in Animals

Aerobic respiration is the primary means of energy production in animals. The process occurs within the mitochondria and involves complete oxidation of glucose in the presence of oxygen.

The overall equation is:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$

The biochemical stages of aerobic respiration in animals are identical to those in plants, namely glycolysis, the Krebs cycle, and oxidative phosphorylation.

Aerobic respiration is highly efficient, producing approximately 36-38 ATP molecules per glucose molecule under optimal conditions.

Anaerobic Respiration in Animals

Anaerobic respiration occurs when oxygen supply becomes insufficient to meet cellular energy demands. In animal tissues, particularly skeletal muscles, pyruvate produced during glycolysis is converted to lactic acid.

The reaction can be represented as:

$C_6H_{12}O_6 \rightarrow 2C_3H_6O_3 + Energy$

This pathway is common during intense physical activities such as sprinting, weightlifting, and vigorous exercise. The accumulation of lactic acid lowers muscle pH and contributes to muscle fatigue and discomfort.

Although anaerobic respiration supplies energy rapidly, it produces only two ATP molecules per glucose molecule, making it significantly less efficient than aerobic respiration.

Respiratory Organs in Animals

Different animal groups have evolved specialized structures for gaseous exchange according to their habitats and physiological requirements.

Mammals

Mammals possess lungs, which provide a large surface area for gas exchange. Oxygen diffuses into the bloodstream through alveolar membranes, while carbon dioxide diffuses out.

Fish

Fish utilize gills for extracting dissolved oxygen from water. Gills contain numerous lamellae that maximize surface area for diffusion.

Insects

Insects possess a tracheal system consisting of spiracles, tracheae, and tracheoles. This system delivers oxygen directly to tissues without reliance on a circulatory transport system.

Amphibians

Amphibians utilize both lungs and moist skin for respiration. Cutaneous respiration is particularly important during aquatic phases of their life cycle.

Earthworms

Earthworms depend entirely on moist skin for gaseous exchange. Oxygen diffuses through the skin into capillaries beneath the epidermis.

Factors Affecting Respiration

1. Temperature – Enzyme-mediated respiratory reactions generally increase with temperature up to an optimum level.
2. Oxygen Availability – Reduced oxygen concentration limits aerobic respiration.
3. Water Content – Dehydration reduces metabolic activity and respiration.
4. Age – Young and actively growing tissues exhibit higher respiratory rates.
5. Activity Level – Increased physical activity elevates energy demand and respiration rate.
6. Availability of Respiratory Substrates – The amount of glucose and other substrates affects ATP production.